What can waves tell us?

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish a signal's travel from the motion of its source.

SUCCESS CRITERIA - I can...

1. Identify what a wave can carry.
2. Separate signal travel time from source speed.
3. Name the measurements or details you need to explain an observation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "wave" and "signal" to explain your answer. Sentence starters are optional.

wave: a disturbance that can carry energy and information

signal: a disturbance that carries information

evidence: observations or measurements that support an idea

01 Warm-up: bring back an earlier idea

Two equal-density spheres have radii r and **3** r. Supplied: area scales by **9**, mass by **27**. **Compare their area per mass**.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: distance = speed × time
t = d/c for light in vacuum.
A signal takes time to travel. Receiving its light does not mean the source traveled with it.

Worked reasoning: A pulse travels **4 m** along a rope in **2 s**, so its average propagation speed is **2 m/s**. A marked rope segment moves locally; it does not ride **4 m** along with the pulse.

Your turn: Sketch three pulse positions and **mark the same rope segment** in each snapshot.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A pulse travels along a rope while each marked rope segment moves mainly up and down. Does the material travel all the way with the pulse?

Sentence starter: I notice ... . This supports ... because ... .



Marzano check - circle: ① ② ③

04 Apply the physics

A wave simulation sends a pulse along a **20 m** rope at **10 m/s**.
1. **Label the one-way path** and **calculate the travel time**.
2. **Explain why a marked piece** of rope does not travel the full **20 m**.

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .



Marzano check - circle: ① ② ③

What can waves tell us?

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student says seeing an asteroid's light tells us exactly where it is "right now." Explain one limitation.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A sample warms by **4°C** after **600 J** is transferred into it.
Explain why the temperature rise and heat transfer need different units.
Label temperatures, samples and relevant energy transfers.

Word help: temperature = a measure of thermal state; it is not an energy amount

Sentence starter: Both … . They differ in … because … .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

07 Exit check

Name one quantity a wave observation could help estimate and one additional piece of information needed.

Word help: estimate = find an approximate value

Sentence starter: I found … [unit]. This value means … in this situation.

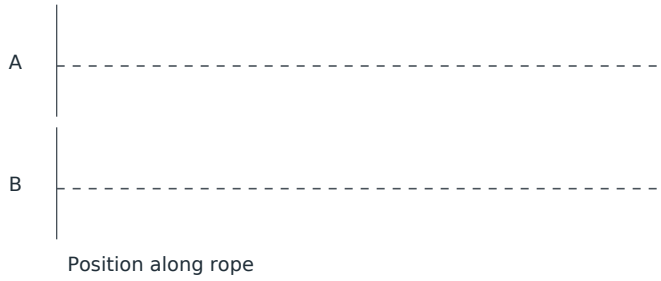


Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. Design a demonstration that distinguishes energy propagation from transport of the entire material.
2. **Name what you would track**.

Sentence starter: My design includes … . It does what is needed by … .



Marzano check - circle: 1 2 3